

Hardware-Aware Representation-Asymmetric Distillation with a Teacher Pool for Tiny Bioacoustic Recognition

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Abstract—Tiny bioacoustic recognition on microcontrollers requires balancing classification accuracy with stringent resource constraints. Existing approaches mainly focus on model compression while paying limited attention to acoustic front-end representations and the practical execution characteristics of target hardware. To address this challenge, we propose a hardware-aware, representation-asymmetric teacher-student framework with a diverse teacher pool. Instead of requiring the deployable student model to learn from the same low-cost representation as the teachers, our framework constructs multiple teachers using diverse network architectures and feature-rich front-end representations. Their soft predictions are aggregated and distilled into a compact student model operating solely on computationally efficient front-end inputs. To further improve deployment performance, the student combines global average pooling and global max pooling. On the BioDCASE-Tiny 2026 Task 3 benchmark, the proposed framework ranked first among all submissions on the ESP32-S3 evaluation track. These results highlight the effectiveness of jointly optimizing acoustic representation, knowledge distillation, and hardware-aware deployment for tiny bioacoustic recognition.

Index Terms—Bioacoustic recognition, knowledge distillation, teacher-student framework, tiny machine learning (TinyML).

1. INTRODUCTION

Audio-based bird species classification is a key task for large-scale biodiversity monitoring. Recent advances in autonomous recording units and tiny machine learning (TinyML) enable this task to run continuously on low-power embedded devices, supporting real-time and privacy-preserving wildlife monitoring [1]. Compared with cloud-based inference, on-device processing reduces communication cost and latency while improving reliability and privacy [2]. However, deploying deep neural networks (DNNs) on microcontrollers remains challenging due to the limited computation, memory, and storage resources available on embedded hardware.

To address this problem, existing TinyML research has mainly focused on designing lightweight neural architectures, model compression, quantization, pruning, and neural architecture search [3]–[6]. These approaches effectively reduce computational complexity and memory consumption, enabling deep learning models to run on resource-constrained devices [3]. Nevertheless, existing methods mainly focus on the deployable model itself and are largely task-agnostic, without addressing the acoustic characteristics specific to bioacoustic signals. In real-world recordings, bird calls typically occupy a small and unevenly distributed region of the time-frequency map, and the features that distinguish acoustically similar species can be sensitive to the time-frequency resolution of the input [7], [8]. These characteristics increase the difficulty of learning discriminative representations under limited model capacity, and addressing this challenge remains an open problem.

In this paper, we propose a DNN-based bird species classification system that can be deployed on tiny hardware. The system uses a MobileNetV2 [3] as backbone, which balances model size and

classification accuracy. At the output stage, we apply max pooling in parallel with average pooling and concatenate the two pooled features, rather than using average pooling alone. The classifier is trained as the student in a teacher-student distillation framework [9]. Unlike standard distillation, we use a pool of teachers with different network architectures and training configurations rather than a single teacher. Averaging predictions across multiple teachers is intended to provide the student with more informative soft labels [10]. Because the teachers are used only during offline training, the distillation framework also allows them to use a higher time-frequency resolution input than the student without adding any inference cost to the student. This asymmetry offers the teachers a potential advantage in extracting discriminative information from the acoustic representation [11], which is then transferred to the student through distillation. The distillation process follows a curriculum learning strategy with two stages [12]. In the first stage, the student is trained on a generated dataset by mixing clean recordings with weak noise. In the second stage, the student is trained on real field recordings from the target deployment distribution. This progression allows the student to first build robust representations under controlled noise conditions before adapting to the acoustic variability present in real recordings.

We evaluate the proposed system on the ESP32-S3-Korvo-2 board using the dataset provided by the BioDCASE Bioacoustics on Tiny Hardware challenge for bird species classification. The proposed system ranks first among all submissions on the official challenge evaluation. On the validation split of the development set, the system reaches 0.744 accuracy, compared with 0.563 for the baseline. On the board, the quantized classifier uses a 78.4 KB TFLite file and 95.9 KB random-access memory (RAM), and runs in 92.7 ms per model inference, compared with 111.4 KB, 202.6 KB, and 330.4 ms for the baseline. Ablation studies further show that both teacher-pool distillation and the two-stage curriculum contribute to this accuracy, and that aggregating the teacher pool yields a more accurate soft label than the best individual teacher.

2. PROPOSED SYSTEM

The proposed system consists of a teacher pool and a student network, in which the student is trained with the teacher-pool soft labels through a two-stage curriculum. The teachers and the student operate on acoustic front ends with different time-frequency resolutions, which allows the teachers to model finer acoustic detail during training without adding any inference cost to the student. Fig. 1 shows an overview of the system, and the following subsections describe the student network, the teacher pool, and the two-stage curriculum distillation procedure in detail.

2.1. Student Network

The student network is the model deployed on the embedded hardware, and its design is therefore driven primarily by deployment constraints such as computation and memory budget. The target hardware in this work is the ESP32-S3-Korvo-2 board. As shown in Fig. 1, the student

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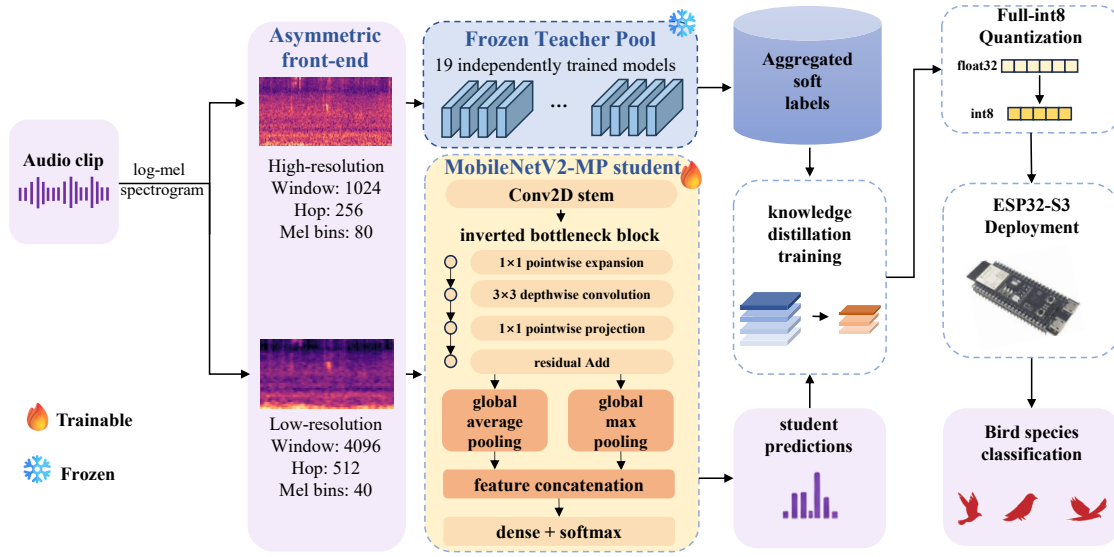


Fig. 1: Overview of the proposed teacher-pool knowledge distillation framework and ESP32-S3 deployment pipeline.

uses a log-mel spectrogram computed with a 4096-sample window, a 512-sample hop, and 40 mel bins.

The backbone of the student network is MobileNetV2 [3], built from a stack of inverted bottleneck blocks with 1×1 pointwise expansion, 3×3 depthwise convolution, and 1×1 pointwise projection, with residual connections where the input and output shapes match. At the pooling stage, we apply max pooling in parallel with original average pooling on the final feature map, and concatenate the two pooled feature vectors before the final dense layer. A bird call typically occupies only a small region of the time-frequency map, so its peak activation can be diluted when averaged with the surrounding background. Max pooling preserves this peak activation regardless of how small the active region is, complementing the global statistics captured by average pooling. We refer to this modified architecture as MobileNetV2-MP.

2.2. Teacher Pool

The teacher pool consists of 19 models trained independently offline. As summarized in Table 1, the pool spans three architectures, a depthwise CNN [13], a ResCNN [14], and a temporal multi-scale CNN [15]. Training configurations further vary in regularization, using dropout [16] and label smoothing [17], in data augmentation, using SpecAugment [18] and MixUp [19], and in random seed. This architectural and training diversity is intended to expose the teachers to different modeling preferences. Each teacher is trained following a two-stage curriculum. In the first stage, the teacher is trained on a simulated training set, formed by mixing real bird recordings with background noise at controlled signal-to-noise ratios (SNRs). In the second stage, the teacher is fine-tuned on a real-world training set of field recordings that match the target deployment distribution. The training strategies listed in Table 1 are applied only in the first stage. All hyperparameters of these strategies were determined through experiments on the validation split from the simulated dataset.

As the teacher models are used only during offline training, their input is not constrained by the target hardware. As shown in Fig. 1, each teacher operates on a log-mel spectrogram computed with a 1024-sample window, a 256-sample hop, and 80 mel bins. The predictions of the 19 teachers are aggregated into a single soft label for distillation.

Table 1: Composition of the 19-model teacher pool.

Architecture	Training Strategy	Count
Depthwise CNN	Standard training (2 seeds)	2
	Label smoothing (0.05, 2 seeds)	2
	Label smoothing (0.075, 0.10)	2
	Dropout (0.20)	1
	SpecAugment (moderate, strong)*	2
	Dropout + label smoothing	1
	MixUp ($\alpha = 0.2$, 3 seeds)	3
	MixUp ($\alpha = 0.4$)	1
	MixUp ($\alpha = 0.1$, $p = 0.5$)	1
	MixUp ($\alpha = 0.2$) + label smoothing	1
	MixUp ($\alpha = 0.2$) + SpecAugment (moderate)	1
ResCNN	Dropout (0.20)	1
Multi-scale CNN	Dropout (0.20)	1
Total		19

*Moderate SpecAugment: 2 frequency masks (max width 10 mel bins), 2 time masks (max width 32 frames). Strong SpecAugment: 2 frequency masks (max width 12 mel bins), 3 time masks (max width 40 frames).

Given an input x , the soft label $\bar{p}_t(x; T)$ is computed by averaging the temperature-scaled softmax outputs of all teachers,

$$\bar{p}_t(x; T) = \frac{1}{M} \sum_{m=1}^M \text{softmax} \left(\frac{z_m(x)}{T} \right). \quad (1)$$

Here, the subscript t indicates that this soft label is obtained from the teacher pool, $z_m(x)$ denotes the logit vector of the m -th teacher, $M = 19$ is the number of teachers, and T is the distillation temperature.

2.3. Two-Stage Curriculum Distillation

The student is trained on the same two training sets used for the teachers, following a two-stage curriculum. At both stages, the student is optimized with a loss that combines the hard label and the teacher-pool soft label $\bar{p}_t(x; T)$,

$$\mathcal{L} = \lambda \mathcal{L}_{\text{CE}}(y, p_s(x; 1)) + (1 - \lambda) T^2 \mathcal{L}_{\text{KL}}(\bar{p}_t(x; T) \| p_s(x; T)), \quad (2)$$

Table 2: Official ranking on the test set of the 2026 BioDCASE Bioacoustics on Tiny Hardware challenge. Only the top two ranked submissions and the official baseline are shown. The proposed system corresponds to submission Hai_task3_1.

Submission	Embedding Model (.tflite)				Embedded Performance on ESP32-S3				
	Accuracy↑	ROC-AUC↑	TFLite Size↓	MACs↓	RAM↓	Setup↓	Preprocessing↓	Model↓	Total↓
Baseline	0.563	0.892	111.4 KB	23.3 M	202.6 KB	4.4 ms	413.6 ms	330.4 ms	743.9 ms
You_PKU_task3_2	0.755	0.958	293.9 KB	40.5 M	2,816.9 KB	2.9 ms	241.1 ms	9,407.6 ms	9,651.6 ms
Hai_task3_1	0.738	0.958	78.4 KB	5.4 M	95.9 KB	7.9 ms	294.7 ms	361.7 ms	664.4 ms

where y is the ground-truth label, $p_s(x; T)$ is the temperature-scaled softmax output of the student, $\mathcal{L}_{CE}$ is the cross-entropy loss, $\mathcal{L}_{KL}$ is the Kullback–Leibler divergence [9], and λ is the weight assigned to the hard label. The two stages share this loss form but differ in the training data and the value of λ .

In the first stage, the student is trained on the simulated training set with $\lambda = 0.3$, so that the soft label dominates the loss and the student mainly learns from the teacher pool. In the second stage, the student is fine-tuned on the real-world training set with $\lambda = 0.9$. This adapts the student to the target deployment distribution, with the hard label dominating the loss while the soft label is retained as a weak regularizer. The values of λ in both stages are determined through pre-experiments.

3. EXPERIMENTAL SETUP

3.1. Datasets

We generate a simulated training set for the first stage of both teacher training and student distillation. The set contains 26,400 clips, with 2,400 clips for each of the 11 classes. Foreground bird vocalizations are mainly collected from Xeno-canto¹. Since the Mallard class has fewer available recordings, additional clips are collected from Tierstimmenarchiv² and iNatSounds 2024. The clips are randomly split into three disjoint groups, accounting for 20%, 50%, and 30% of the set. No additional noise is added to the first group. The clips in the second and third groups are mixed with a randomly selected background noise segment from Freefield1010³ and WarblrB10k [20], at an SNR randomly sampled from [3, 12] dB for the second group and from [−6, 0] dB for the third group.

We use the real-world development set⁴ provided by the BioDCASE Bioacoustics on Tiny Hardware challenge for the second stage of both teacher training and student distillation. The development set contains 2,750 samples in total. We randomly split the set into a training split of 2,200 samples and a validation split of 550 samples, corresponding to approximately 200 training samples and 50 validation samples per class. The validation split is held out from training and is used only for evaluation. All audio clips used in this work are 3-second segments sampled at 24 kHz.

3.2. Implementation and Training Details

The teachers and the student follow the same training configuration. In the first stage, training runs for up to 80 epochs with an early stopping patience of 12 epochs, using the Adam optimizer with a learning rate of 10^{-3} and a batch size of 32. In the second stage, fine-tuning runs for up to 40 epochs with a learning rate of 10^{-4} . The distillation temperature is set to $T = 2.0$.

¹<https://xeno-canto.org/>

²<https://suche.tierstimmenarchiv.de>

³https://dcase-repo.github.io/dcase_datalist/datasets/sounds/qmul_freefield1010.html

⁴<https://zenodo.org/records/19453065>

Table 3: Classification performance and deployment cost on the validation split of the real-world development set.

Method	Accuracy ↑	ROC-AUC ↑	Size ↓	MACs ↓	RAM ↓	Latency ↓
Baseline	0.563	0.892	111.4 KB	23.3 M	202.6 KB	330.4 ms
Ours	0.744	0.947	78.4 KB	5.4 M	95.9 KB	92.7 ms*

*To explore the achievable on-device model latency, our evaluation on the validation split enables the optimized ESP-NN kernels for the Conv2D operators. The official evaluation toolchain instead uses the default kernel dispatch, and a substantially higher model latency of 361.7 ms is reported in Table 2 for the proposed system.

3.3. Deployment and Evaluation

After training, the student model is converted to a fully int8 TensorFlow Lite model using post-training quantization [4], [6], with a calibration set of 1,024 samples randomly selected from the training split of the real-world development set. The quantized student model is compiled with the Espressif IoT Development Framework (ESP-IDF) v5.4.4⁵ and evaluated on an ESP32-S3-Korvo-2 board running at 240 MHz, with 8 MB PSRAM and 16 MB Flash. Inference is executed with TensorFlow Lite Micro [6], using ESP-NN kernels⁶ when available.

Classification performance is measured by accuracy and the area under the receiver operating characteristic curve (ROC-AUC). Deployment cost is measured by the TFLite model size, the multiply-accumulate operations (MACs), the RAM usage, and the model inference latency. The RAM usage refers to the memory occupied during inference, and model latency is measured as the wall-clock time for a single forward pass on the board.

4. RESULTS AND ANALYSIS

We evaluate the quantized student network deployed on the ESP32-S3-Korvo-2 board, on a classification task with 11 classes, corresponding to 10 target bird species and one background class. Unless otherwise noted, all reported results are computed on the validation split of the real-world development set.

4.1. Overall Performance

Table 2 reports the official ranking of the Bioacoustics on Tiny Hardware challenge. Among all submissions, the proposed system is the official winning submission, achieving 0.738 accuracy and 0.958 ROC-AUC. Compared with the runner-up submission (You_PKU_task3_2), the proposed system reduces the TFLite model size from 293.9 KB to 78.4 KB, the MACs from 40.5 M to 5.4 M, the RAM from 2,816.9 KB to 95.9 KB, and total on-board latency from 9,651.6 ms to 664.4 ms, at comparable classification accuracy and ROC-AUC. These results indicate that the proposed system achieves the best trade-off between accuracy and deployment cost among all submissions.

Table 3 compares the proposed system with the baseline on the validation split. The proposed system improves accuracy from 0.563

⁵<https://github.com/espressif/esp-idf>

⁶<https://github.com/espressif/esp-nn>

Table 4: Validation accuracy of the student model across the two training stages, with and without teacher-pool distillation.

Distillation	Label	Stage I	Stage II
Without	Hard label	0.607	0.718
With	Hard label + soft label	0.656	0.744

Table 5: Validation accuracy of the single teacher and the 19-teacher pool across the two training stages.

Configuration	Stage I	Stage II
Single teacher (highest validation accuracy)	–	0.769
Teacher pool (19 teachers)	0.736	0.798

to 0.744 and ROC-AUC from 0.892 to 0.947. On the ESP32-S3-Korvo-2 board, it reduces the TFLite model size from 111.4 KB to 78.4 KB, the RAM from 202.6 KB to 95.9 KB, and the model latency from 330.4 ms to 92.7 ms. The proposed system therefore improves classification accuracy while reducing deployment cost across all measured metrics.

4.2. Ablation Study

This section examines the two design choices central to the proposed system, teacher-pool distillation and the two-stage curriculum. Table 4 isolates the contribution of teacher-pool distillation across the two training stages. With only the hard label, accuracy improves from 0.607 to 0.718 across the two stages. With the hard label and the teacher-pool soft label combined, accuracy improves from 0.656 to 0.744. At both stages, adding the soft label outperforms training with the hard label alone. These results show that the two-stage curriculum and teacher-pool distillation each contributes to the accuracy of the student model, and the highest accuracy is reached when both are applied together. Table 5 presents an ablation on teacher-pool aggregation. At Stage II, the teacher pool reaches an accuracy of 0.798, compared with 0.769 for the single teacher. The teacher pool also improves from 0.736 at Stage I to 0.798 at Stage II, following the same two-stage trend observed for the student in Table 4. These results show that aggregating predictions across the 19 teachers produces more accurate soft labels than the best individual teacher alone, providing the basis for the improvement in student accuracy achieved by teacher-pool distillation. Comparing Table 5 with Table 4, the teacher pool outperforms the student at both stages, and this gap narrows as training progresses. The advantage of the teacher pool may partly reflect the higher time-frequency resolution available to the teachers during offline training, which the student cannot adopt without increasing memory and computation cost.

4.3. Per-Class Error Analysis

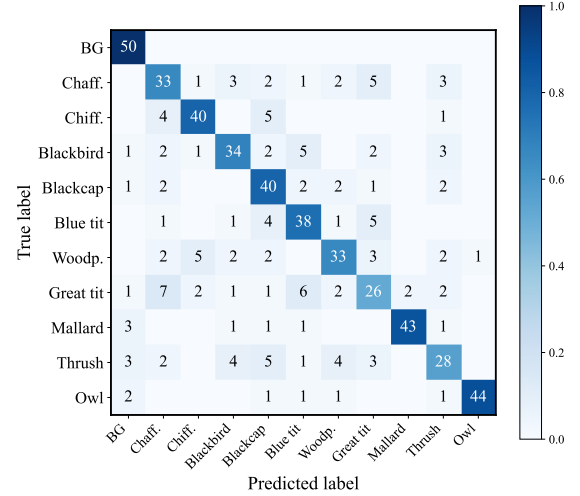
Figure 2 shows the confusion matrix of the student model on the validation split. The student classifies all background clips correctly, and achieves high recall on Owl and Mallard. In contrast, Great Tit and Thrush are the most difficult classes, with recall dropping to 26/50 and 28/50, respectively. Great Tit is mainly confused with Chaffinch and Blue Tit, while Thrush errors are distributed across Blackcap, Blackbird, and Great Spotted Woodpecker. These confusions are concentrated in a small number of species pairs rather than spread evenly across all classes, which is consistent with confusion between acoustically similar species rather than a general limitation of the model.

4.4. Operator-Level Runtime Analysis

Table 6 breaks down the 92.7 ms inference latency reported in Table 3 by operator type. Conv2D and DepthwiseConv2D together account

Table 6: Operator-level runtime breakdown of the deployed student on the ESP32-S3-Korvo-2 board.

Operations	Runtime	Share
Conv2D	46.8 ms	50.4%
DepthwiseConv2D	29.1 ms	31.4%
Residual addition	4.7 ms	5.1%
Global average pooling	5.9 ms	6.3%
Global max pooling	5.4 ms	5.9%
Dense layer	0.6 ms	0.7%
Softmax	0.2 ms	0.2%
Total	92.7 ms	100.0%

**Fig. 2:** Validation confusion matrix (accuracy: 0.744). Numbers: sample counts; colors: row-normalized proportions.

for 81.8% of the total latency, consistent with these operators forming the bulk of the MobileNetV2-MP backbone. The remaining operators, including residual addition, the two pooling branches, the dense layer, and softmax, each contribute a small fraction of the total latency. This breakdown also indicates that further latency reduction should primarily target the convolutional operators, as optimizing the remaining operators would yield only marginal gains.

5. CONCLUSION

This paper presented a bird species classification system for deployment on tiny hardware, trained with teacher-pool distillation and a two-stage curriculum. The system uses a pool of 19 teachers spanning diverse architectures and training strategies, whose predictions are aggregated into a soft label for distillation, following a two-stage curriculum that progresses from a simulated training set to real-world field recordings. On the BioDCASE Bioacoustics on Tiny Hardware challenge, the proposed system ranked first among all submissions on the official evaluation set. On the validation split of the development set, it reaches 0.744 accuracy, compared with 0.563 for the baseline, while reducing the quantized model size, RAM, and model inference latency. Ablation studies further show that teacher-pool distillation benefits both the quality of the soft label and the accuracy of the distilled student. These results indicate that combining a diverse teacher pool with a two-stage curriculum is an effective strategy for training compact bird species classifiers under the resource constraints of tiny hardware, and the approach may extend to other embedded acoustic intelligence applications.

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